OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 36 OF 55

INTELLIGENT WASTE SEGREGATION AND CLASSIFICATION USING COMPUTER VISION AND DEEP LEARNING

Domain: Environmental AI & Robotic Sorting | **Module:** project_36_waste_classification

OFFICIAL ABSTRACT

To enhance segregation accuracy, the system incorporates transfer learning using pre-trained architectures, enabling robust feature extraction even with limited training samples. The model is trained on a comprehensive dataset of common household and industrial waste items, achieving high accuracy in classification and localization. The output of the deep learning model is integrated with robotic actuators or pneumatic sorters to physically separate the classified items in real-time. Experimental results demonstrate that the proposed system provides high sorting throughput and classification accuracy compared to manual processes. Implementing this automated system in recycling centers can significantly increase recycling efficiency, reduce landfill dependency, and promote circular economy practices.

📌 Project Summary & System Specifications

Project Title	INTELLIGENT WASTE SEGREGATION AND CLASSIFICATION USING COMPUTER VISION AND DEEP LEARNING
Specialization / Category	Environmental AI & Robotic Sorting
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5036`
Local Implementation	`project_36_waste_classification`